Design and Construction of RFID Environment to Track Shuttle Cab Ride and Access Control

Final Project Report



VIT®
Vellore Institute of Technology
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CSE2006 Microprocessors and Interfacing

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ABSTRACT-

Thousands of students use shuttle cab in VIT on daily basis. This generally involves students to wait in line in shuttle cab to pay for each ride. The fare is Rs 15, a small amount, but is still a very tedious process to pay every time. Also, it is difficult for cab riders to keep track of number of students who entered the cab and actually paid for the ride. We plan to a system using RFID tag and scanner to detect entry of a student in the shuttle cab. The RFID tag would be located inside ID card of students and the scanner would be near the doors to mark the entry or exit of student. The data collected over a time (Eg. No of rides, time of rides, frequency etc.) period will be available for the user to see. The system will also aggregate the fare of all the rides over a monthly time and the user will be able to pay the total amount of money without going through the trouble of finding change of Rs 15 all the time. The details about the rides, as described earlier will be available to the user using a small mobile application available on Android platform for free. The use of micro controller, RFID components and some other electronics coupled together will help to achieve our aim. From this project we hope to build an easy and reliable tracking system for students to use for their convenience and expand our learning of the subject to help the society in similar way in the future.

INTRODUCTION-

RFID has been an emerging technology in recent years. RFID technology can be effectively employed in number of applications due to its penchant for efficiency. As for its application, it's been a widespread tool for both tracking the transit transports. A fundamental system of RFID consists of two primary components: The reader circuit and tag, details of which are discussed later. The usage of RFID has a great advantage as it is considered to be an integral part of IoT. IoT refers to a global network infrastructure, linking physical and virtual objects through exploitation of data capture and communication capabilities. The proposed system mainly acts to bring out the consistency among various bus agencies that will conclude in uniform access of passengers in daily rides through an automated server being updated every single time the passengers travel by carrying the RFID based tickets.

Literature Review-

S. no.	Paper title	Result	Observation	Author
1.	Bus tracking and monitoring using RFID	Implemented low price Bus monitoring scheme that enables the commuters to understand exactly where the bus is located.	For all the paths, it can be introduced. This scheme includes RFID reader with low coverage frequency; it can be achieved using RFID reader with elevated coverage frequency.	P. A. Kamble and R. A. Vatti
2.	Analysis of an automated bus tracking system for metropolitan using IOT	Various technology and methods are used to implement the bus tracking system and location finding technique.	We can create a system for future enhancement that can track, notify and track the car by incorporating the RFID that can identify the items with GSM to enhance system efficiency	K. G. Subadra, J. M. Begum and H. Dhivya
3.	A new generation of smart UHF RFID reader	A novel fresh generation of fixed reader UHF RFID is unleashed in this document. The reader is intended with very elevated sensitivity of receiving discrete	Combined with a strong computing CPU, this reader offers a platform for high-speed computing capacity and data retention, making it possible to	P. Y. Lau, Chu Qingxin and Wu Yueshan

		components. It can be commonly embraced for future intelligent RFID system apps with its unique characteristics and capacities.	range, locate, and detect movement and other apps.	

OBJECTIVE-

- Expertise in construction, lean construction, ICT in construction, i.e. operation, consultancy, systems
- development, implementation
- Expertise in ICT-development, systems and applications
- Expertise in logistics, supply chain management and lean production.
- Expertise in product ID, standards, bar-coding and RFID-technology
- Expertise within change management processes, technology foresight, and vision management.
- Networks within construction, supply chain management, ICT, product ID, facilities management, asset management.

Methodology and Software used-

Our method consists of three parts

1. Detecting RFID tag
2. Processing of data by Arduino
3. Simulation through Keil µVision and Circuit Representation in Proteus

Hardware Description-

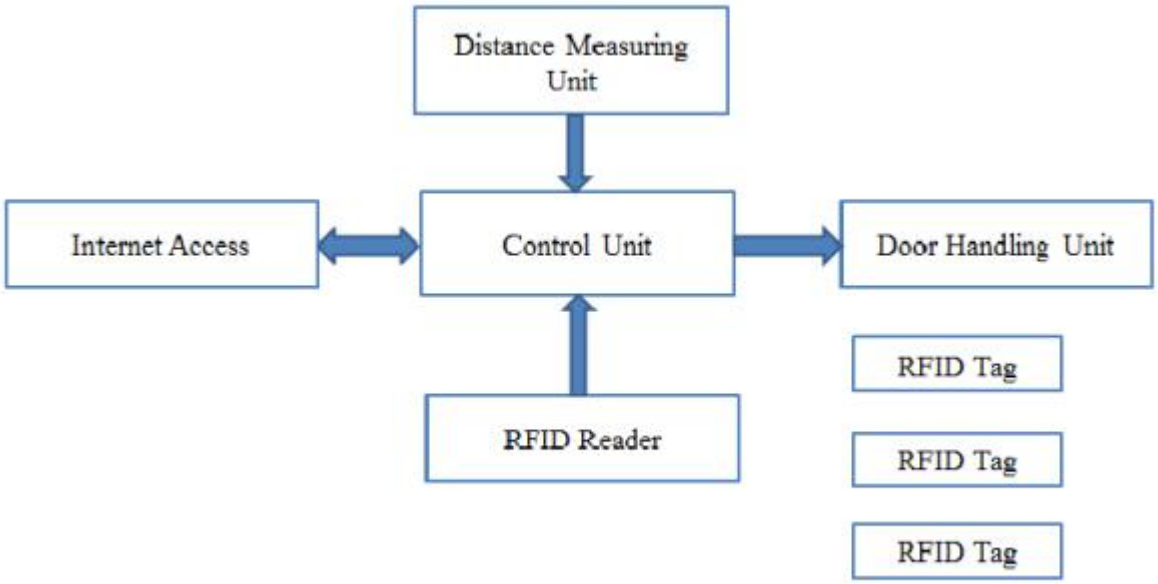
1. RFID Tag

RFID tags are the components which are utilized for the purpose of identification. The tag has a microchip and an integrated antenna. Corresponding to each tag, the microchip contains unique digital data. The most significant feature of RFID tag is the uniqueness exhibited by each of them. When the tag is read, digital data in the chip is send through radio frequency interference technique. These cards may be of different size and range. Passive tags with no batteries have long life and shorter reading range and are ideal for mass identification process giving the advantage of low cost. Cards shaped S50 RFID cards are ideal for the usage here.

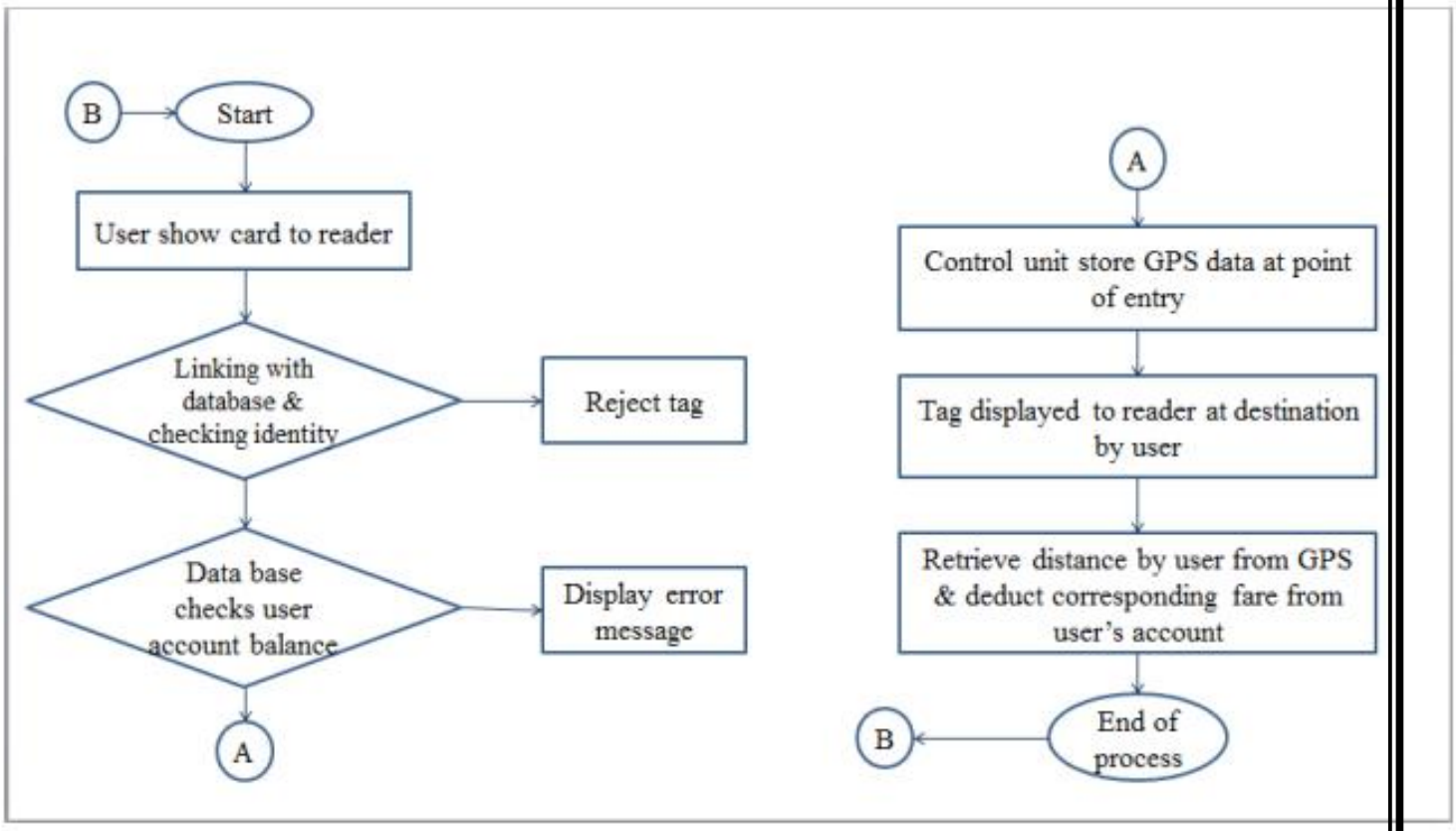
1. RFID Reader

The unique digital data of tag is decoded with the use of RFID reader. The RFID reader transmits an electromagnetic wave which is input to the tag. RFID tag is energized due to these electromagnetic waves hence resulting in the production of a confined magnetic field, which has an interference pattern. This interference pattern which when read by a RFID reader would produce the unique number assigned to the RFID tag and thus the address of the tag is obtained. It should be noted that the address defers from each RFID tag as they are provided by EPCglobal and hence it offers complete resistance to duplication. Here we use MFRC522 reader for reading passive tags.

WORKING MECHANISM-



An RFID system consists of a tag, basically a microchip with an antenna and an interrogator or reader with an antenna. Most RFID tags contain at least two parts, One is an integrated circuit for storing and processing information, modulating and demodulating a radio-frequency (RF) signal, and other specialized functions. The second is an antenna for receiving and transmitting the signal. The passenger entering the desired bus should display/place the RFID tag in front of the reader. When the tag is placed before the reader circuit, the tag gets energized and the reader reads the unique digital data behind it. Hence the tag reveals relevant information to the reader circuit. The reader circuit stores this information in its internal memory temporarily and also links to the common database system which has all the details of the particular passenger. It verifies the identity of the person. Having linked with the common database system, it also checks if the passenger has sufficient balance in his account for the travel. Only a person with minimum account balance and verified identity will be able to travel in the bus which assures secure and reliable transportation system. After each travel, the individual bus unitsupdate database by transferring credit to the corresponding account and also the information can be found in the main database. Cross checking of all those information will allow better monitoring, transparency and thus reducing corruption.



Detecting RFID tag-

We will use passive RFID system which will constitute of an RFID reader, RFID tags and a microcontroller. Passive tag will wait for signal from an RFID sensor. The reader will send energy to an antenna which will convert that energy into a radio frequency wave that is sent into the read zone. If there is any student within the read zone of RFID reader then antenna of RFID tag will draw energy from sent radio frequency wave. This energy will move from RFID's antenna to its integrated circuit and powers the chip which will generate a signal back to RFID reader. This change in radio frequency wave due to loaded information is detected by radar and radar will interpret this information. Read range of passive tag will be of about 1-10 centimeters which will be having a low frequency of 125-134 KHZ. This tag is also not affected by water.



Processing Data-

After getting data from RFID tag we need to process the data in its actual form which will be then sent to the sever. To do the processing we will use Arduino which will be connected to RFID sensor in physical manner. Now Arduino will be responsible for making required operations like authenticating registration number. Arduino is also connected to internet so that it can process data from internet server. Registration number of every student is stored on the server in a database.After receiving registration number from the user's RFID tag, Arduino will match itself to total fare column of respective ID. Once registration number is found total fare will be incremented by fare of one ride.

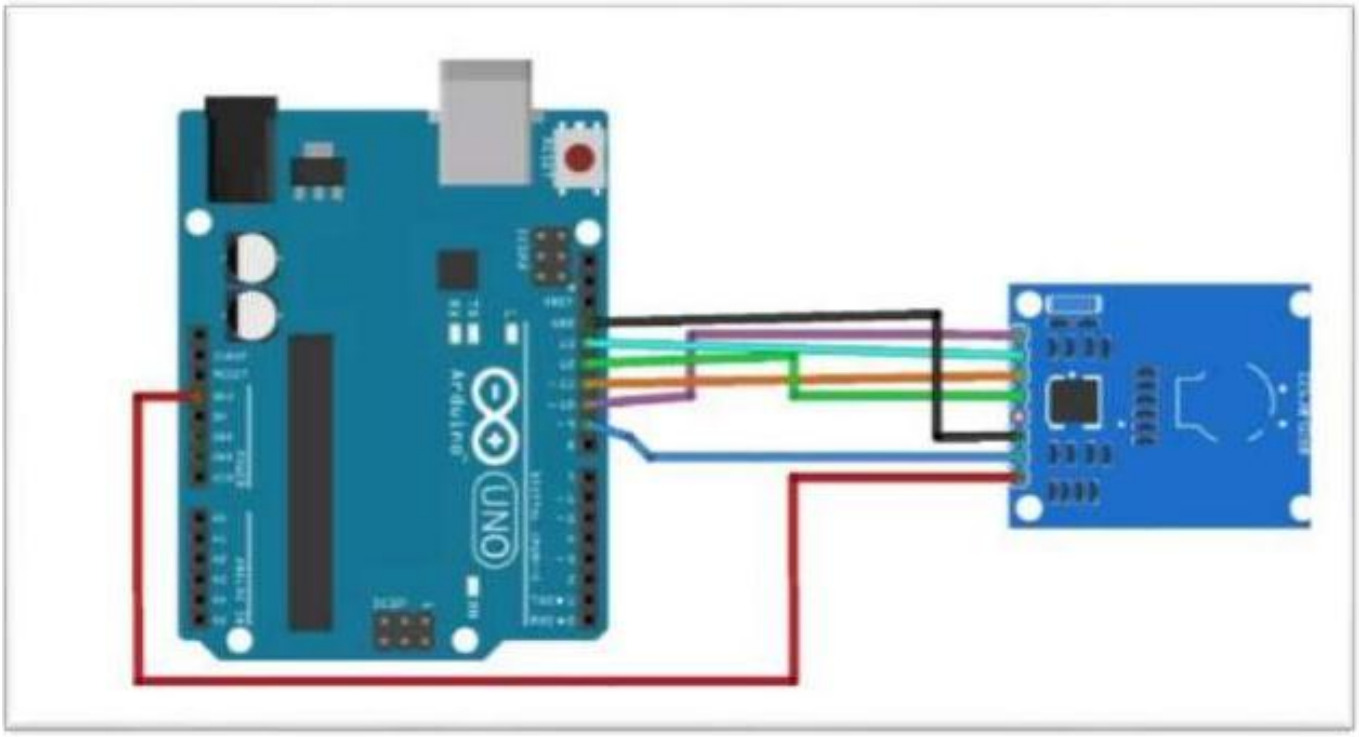


GUI for Payment Gateway-

As recommended by us a user can pay total fare on monthly basis, however final decision will be of user about how to pay. To make payment feasible we will provide a user interface as an android application. To make the GUI of android application we will use android studio which is based on java.



Architecture diagram and table-



Signal	MFRC522 Reader/PCD Pin	Arduino Uno Pin
RST/Reset	RST	9
SPI SS	SDA	10
SPI MOSI	MOSI	11
SPI MISO	MISO	12
SPI SCK	SCK	13

IMPLEMENTATION-

Code:

```
RS EQU P2.7
RW EQU P2.6
E  EQU P2.5
FLG EQU 40H
SEL EQU 41H

ORG 000H
CLR P2.0
MOV TMOD,#00100001B
MOV TH1,#253D
MOV SCON,#50H
SETB TR1
ACALL LCD_INIT
ACALL LINE2

MAIN:ACALL LCD_INIT
    MOV DPTR,#TEXT1
        ACALL LCD_OUT
        ACALL LINE2
        MOV DPTR,#TEXT2
        ACALL LCD_OUT
        ACALL READ_TAG
        CLR REN
        ACALL LINE2
        ACALL CHECK_PASSWORD
```

```
        ACALL DELAY1
        SETB REN
        SJMP MAIN


LCD_INIT: MOV DPTR,#INIT_COMMANDS
         SETB SEL
                ACALL LCD_OUT
                CLR SEL
                RET

LCD_OUT:  MOV R1,#14D
PO:       CLR A
          MOVC A,@A+DPTR
                JZ EXIT
                INC DPTR
                JB SEL,CMD
                ACALL DATA_WRITE
                DJNZ R1,PO
                SJMP LCD_OUT
CMD:     ACALL CMD_WRITE
          SJMP LCD_OUT
EXIT:    RET

LINE2:MOV A,#0C0H
    ACALL CMD_WRITE
        RET

 READ_TAG:  MOV R0,#3D
           MOV R1,#3FH
          WAIT:JNB RI,WAIT
```

```
        MOV A,SBUF
            MOV @R1,A
            INC R1
            CLR RI
            DJNZ RO,WAIT
            RET


CHECK_PASSWORD: CLR FLG
        MOV R2,#3D
                MOV R1,#3FH
                MOV DPTR,#PASS1
    REPEAT: CLR A
            MOVC A,@A+DPTR
                XRL A,@R1
                JNZ CHECKNEXT
                INC R1
                INC DPTR
                DJNZ R2, REPEAT
                ACALL LINE2
                MOV DPTR,#TEXT3
                ACALL LCD_OUT
                SETB P2.0
                RET

CHECKNEXT: JB FLG,FAIL
        MOV R2,#3D
                MOV R1,#3FH
                MOV DPTR,#PASS2
                SETB FLG
                SJMP REPEAT
```

```
FAIL:ACALL LINE2
    MOV DPTR,#TEXT4
        ACALL LCD_OUT
        CLR P2.0
        RET



DELAY1:MOV R3,#46D
BACK:  MOV TH0,#00000000B
       MOV TL0,#00000000B
          SETB TR0
HERE1: JNB TF0,HERE1
       CLR TR0
          CLR TF0
          DJNZ R3,BACK
          RET


CMD_WRITE: MOV P0,A
   CLR RS
       CLR RW
       SETB E
       CLR E
       ACALL DELAY
       RET

DATA_WRITE:MOV SBUF,A
HERE2: JNB TI,HERE2
```

```
   CLR TI
       SETB RS
       CLR RW
       SETB E
       CLR E
       ACALL DELAY
       RET

DELAY: CLR E
   CLR RS
       SETB RW
       MOV P0,#0FFh
       SETB E
       MOV A,P0
       JB ACC.7,DELAY
       CLR E
       CLR RW
       RET

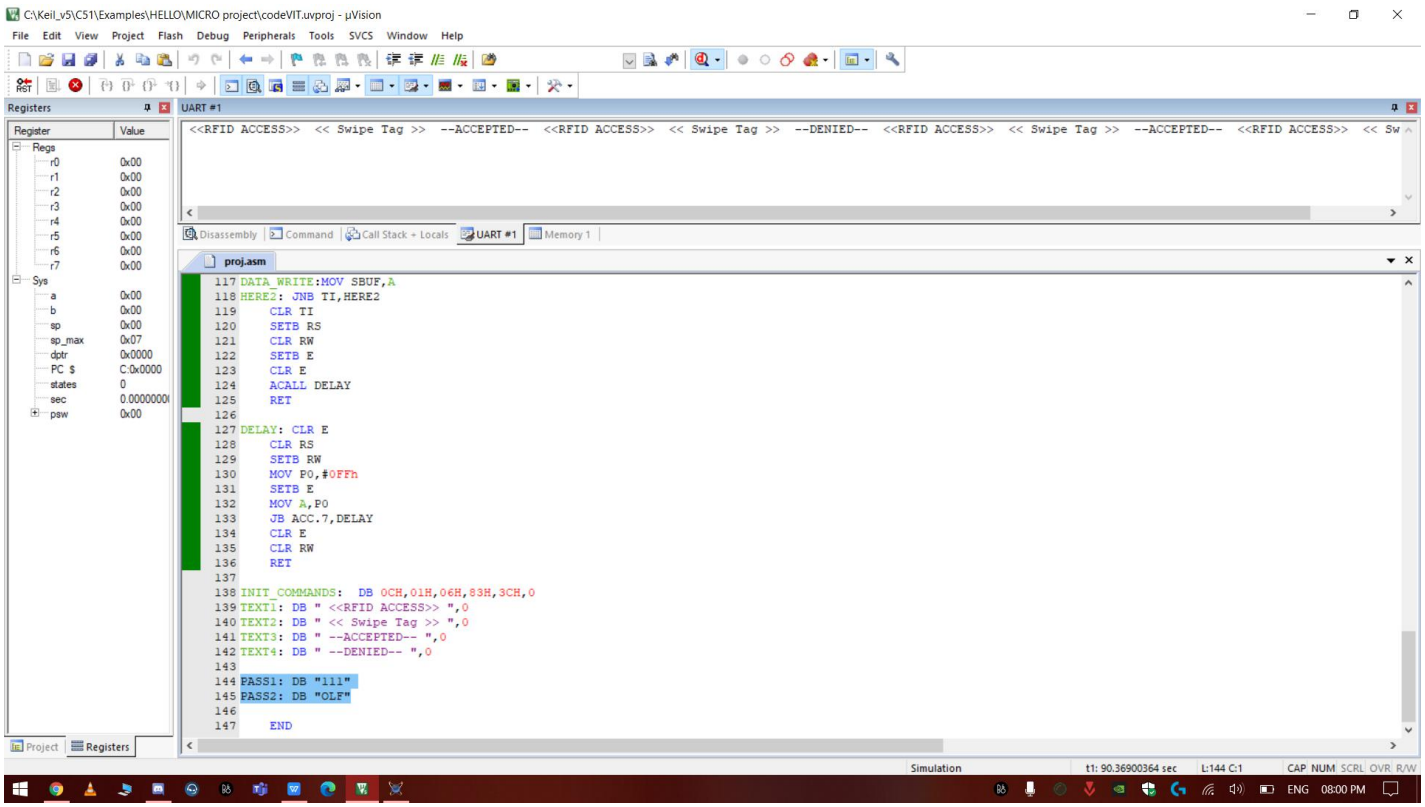
INIT_COMMANDS:  DB 0CH,01H,06H,83H,3CH,0
TEXT1: DB " <<RFID ACCESS>> ",0
TEXT2: DB " << Swipe Tag >> ",0
TEXT3: DB " --ACCEPTED-- ",0
TEXT4: DB " --DENIED-- ",0

PASS1: DB "111"
PASS2: DB "0LF"

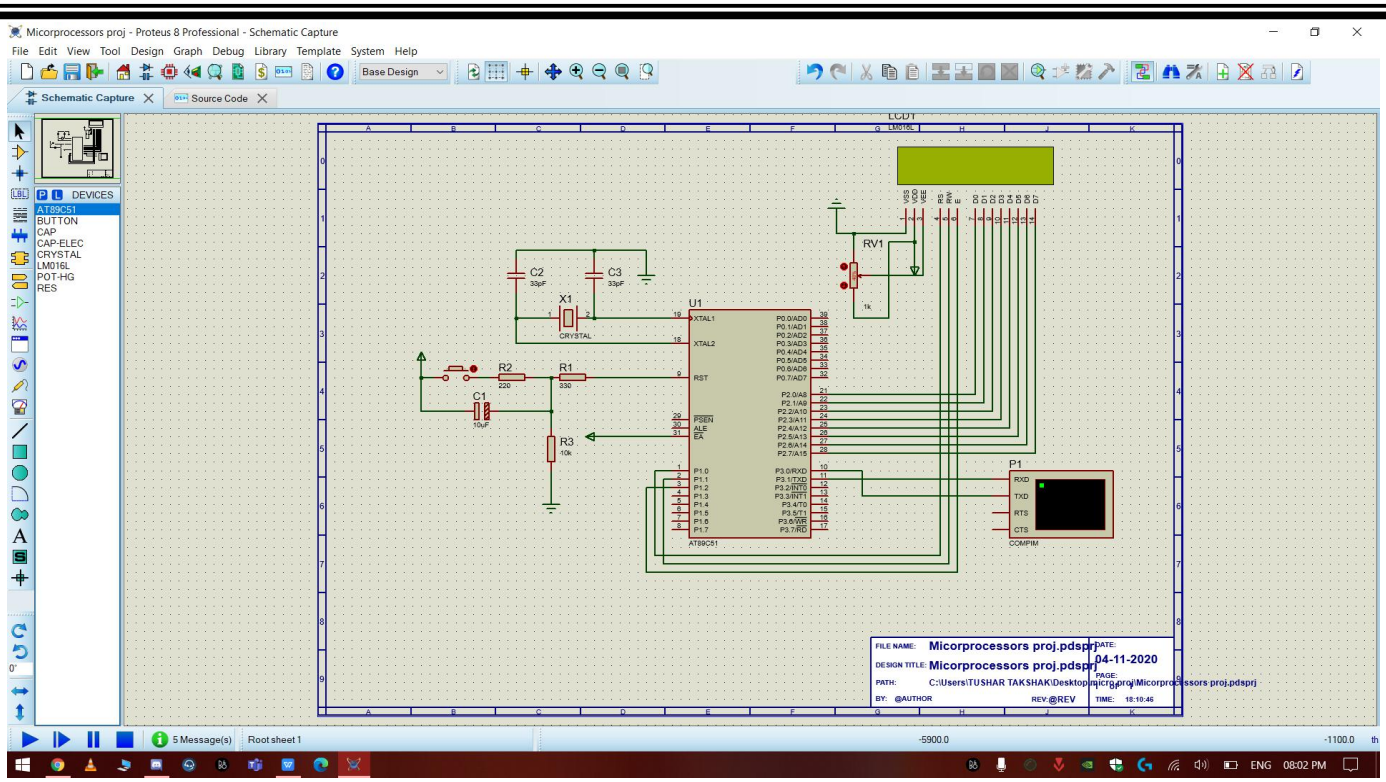
       END
```

OUTPUT-

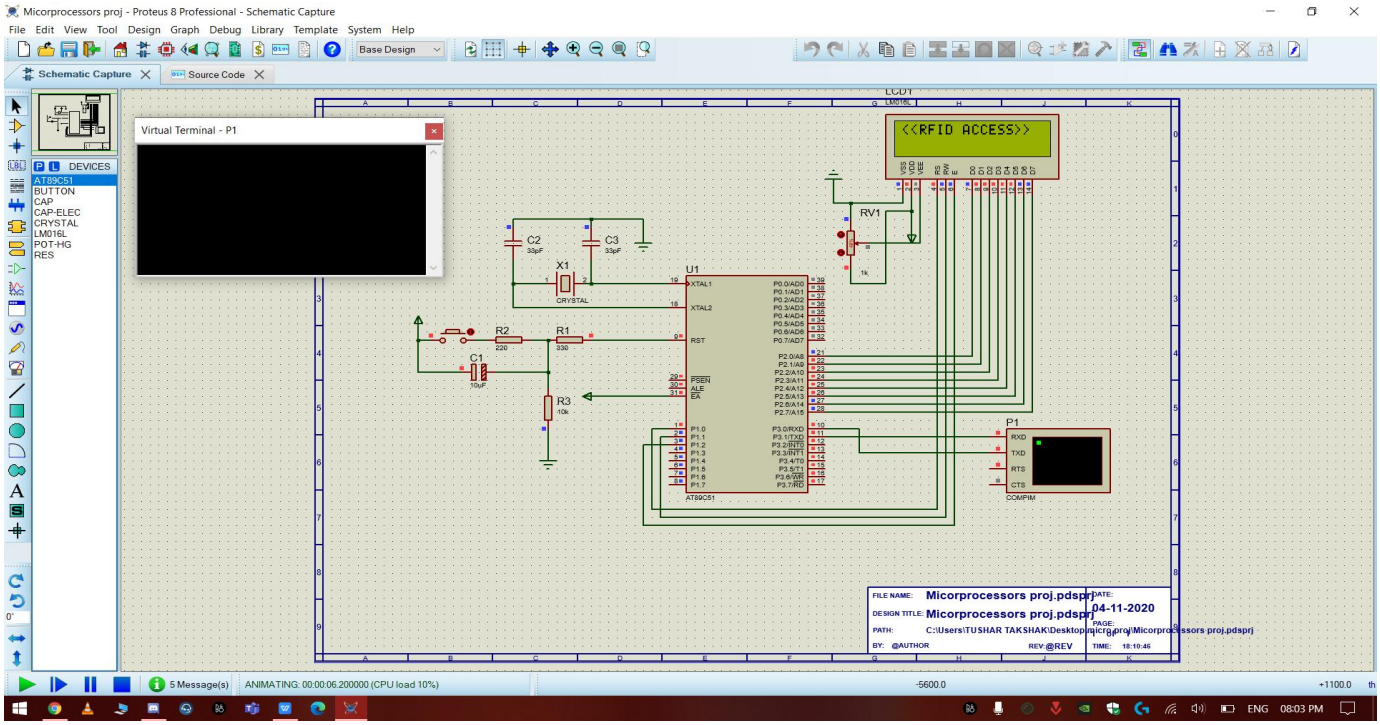
In the UART window the correct pass shows --ACCEPTED-- and the wrong pass shows --DENIED--



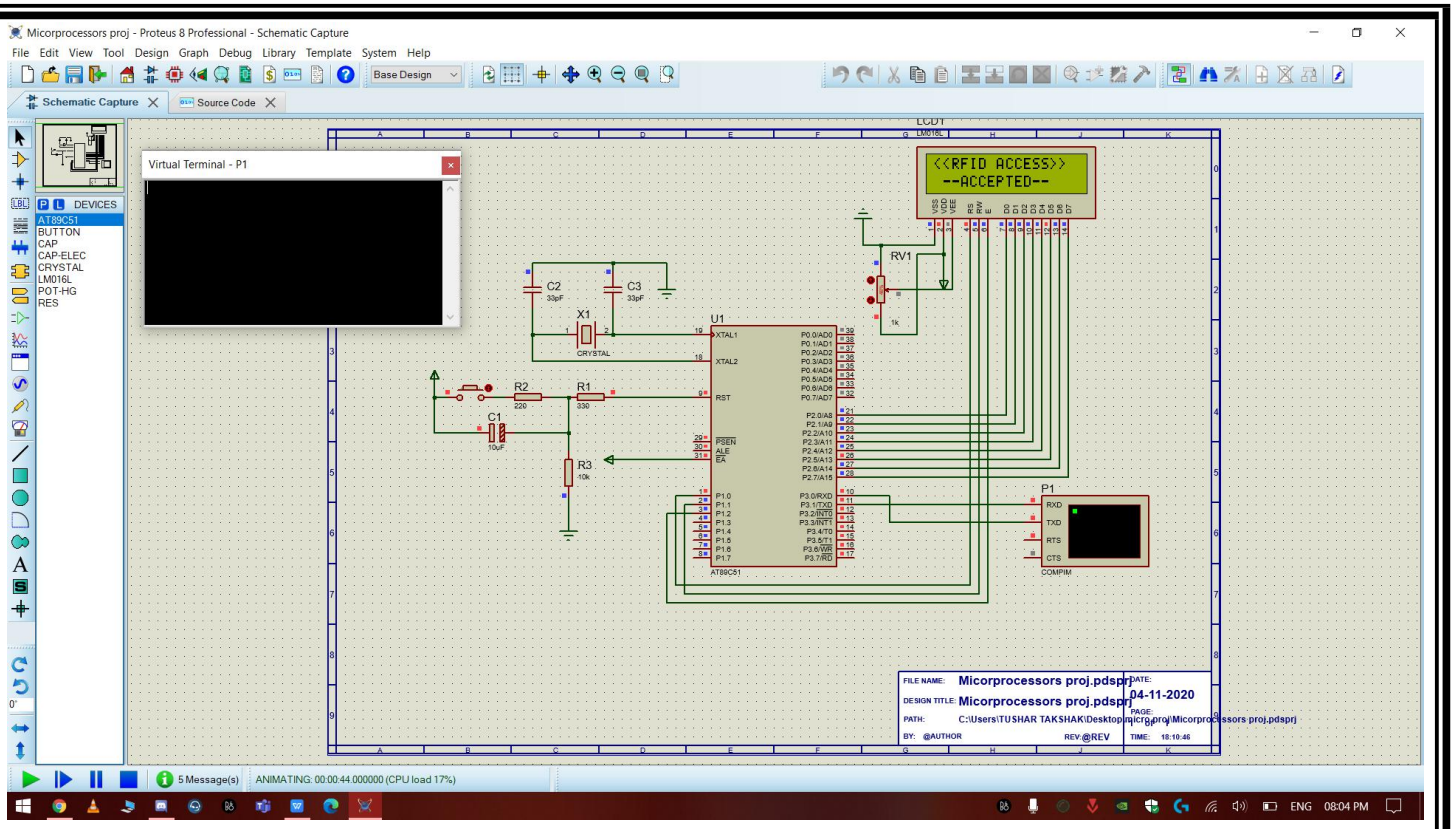
Circuit in Proteus with a hex file as program file-



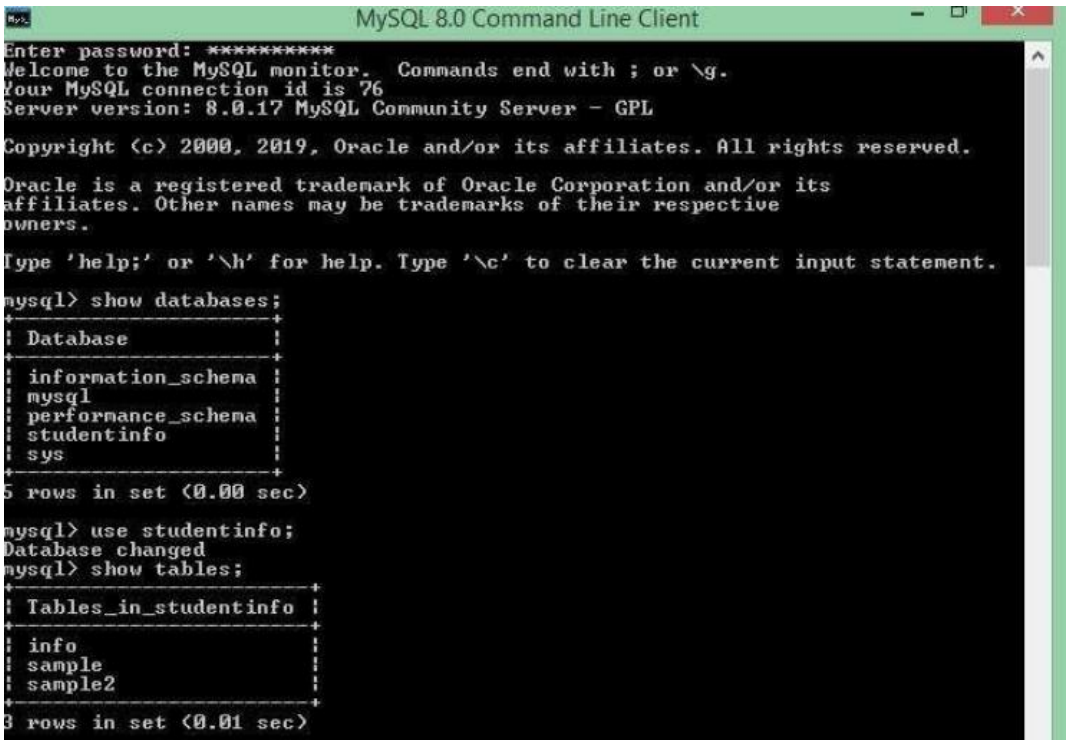
Before entering pass:

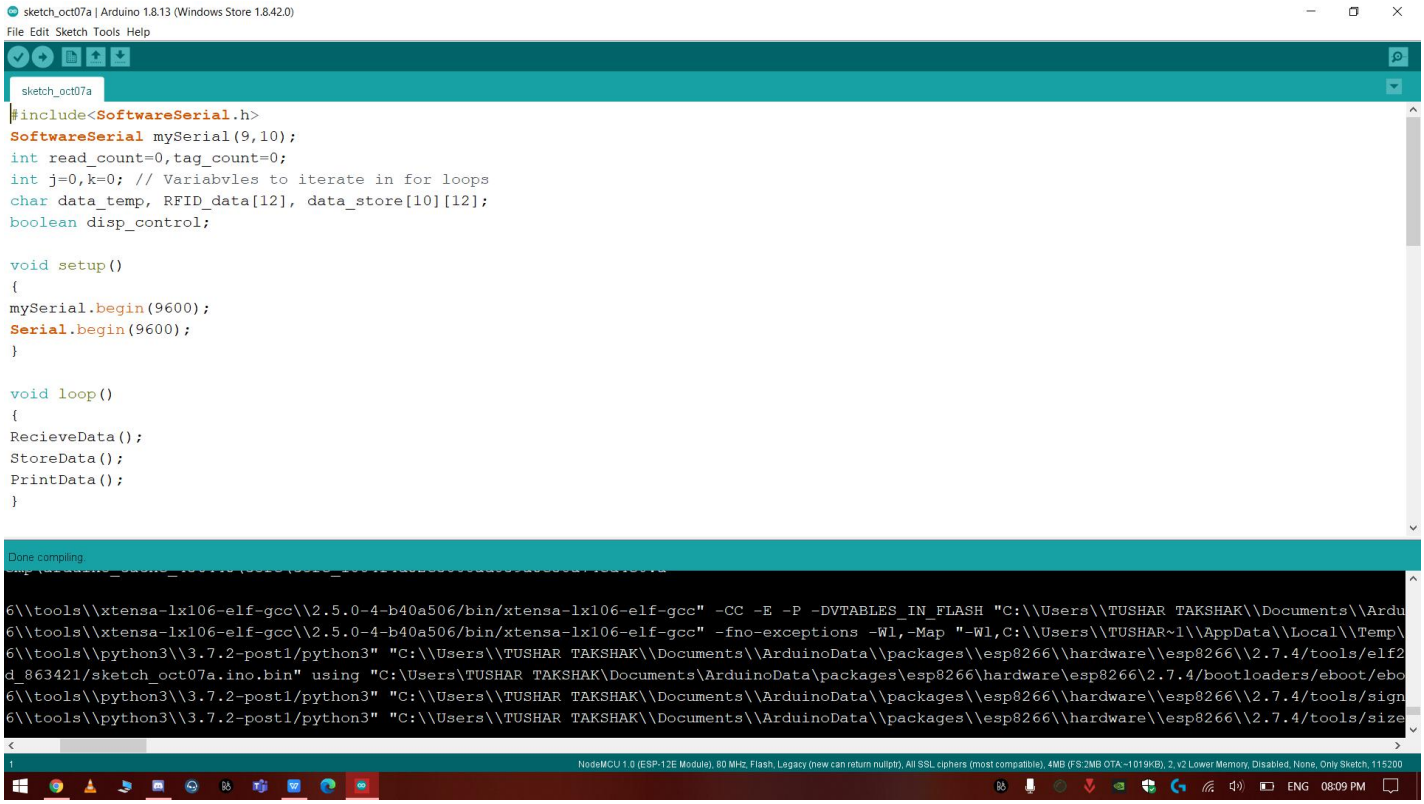


After entering correct pass:


Micorprocessors proj - Proteus 8 Professional - Schematic Capture
File Edit View Tool Design Graph Debug Library Template System Help
Schematic Capture
Source Code
Virtual Terminal - P1
<<RFID ACCESS>>
--ACCEPTED--
LCD1
LM016L
RV1
C2
C3
X1
CRYSTAL
U1
AT89C51
R1
R2
R3
C1
P1
COMPIM
FILE NAME: Micorprocessors proj.pdsprj
DESIGN TITLE: Micorprocessors proj.pdsprj
PATH: C:\Users\TUSHAR TAKSHAK\Desktop\micro proj\Micorprocessors proj.pdsprj
BY: @AUTHOR
REV:@REV
DATE: 04-11-2020
TIME: 10:10:46
5 Message(s)
ANIMATING: 00:00:44.000000 (CPU load 17%)
ENG 08:04 PM

Database:


MySQL 8.0 Command Line Client
Enter password: **********
Welcome to the MySQL monitor. Commands end with ; or \g.
Your MySQL connection id is 76
Server version: 8.0.17 MySQL Community Server - GPL
Copyright (c) 2000, 2019, Oracle and/or its affiliates. All rights reserved.
Oracle is a registered trademark of Oracle Corporation and/or its
affiliates. Other names may be trademarks of their respective
owners.
Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.
mysql> show databases;
+--------------------+
| Database |
+--------------------+
| information_schema |
| mysql |
| performance_schema |
| studentinfo |
| sys |
+--------------------+
5 rows in set (0.00 sec)
mysql> use studentinfo;
Database changed
mysql> show tables;
+----------------------+
| Tables_in_studentinfo |
+----------------------+
| info |
| sample |
| sample2 |
+----------------------+
3 rows in set (0.01 sec)



Possible Future Upgrades-

1. We can add a secure encryption system to transfer data between servers and Arduino.

2. We can include a biometrics system that helps us insecure authentication of the user. Because of no authentication system, anyone can come with someone else ID and enter a shuttle can, which will lead to an increment of fare for the wrong user.

Results-

1: The system is fully automated, reliable, transparent, and convenient. The cards being reusable, they are much more convenient compared to the paper-based ticketing system.

2: Also fare calculation is done through the internet, fare is crystal clear and provides no room for confusion.

3: Cash is no longer necessary, contactless smart cards can be loaded with large amounts of money, passengers no longer need to carry the correct change

Possible Future of this Project-

Imagine sign board on our roads having RFID tags and a receiver in bus. These signals can work without power for a very long time unless you destroy them physically. With respect to GPS which constantly requires a high power source to operate, RFID is a better alternative. This system will be way better than GPS based on strength of the signal needed by both of them. RFID can easily operate in a low signal area. We can easily track each bus and can give a real-time position of the bus. RFID is also economic and easy to operate.

Conclusion-

We implemented a shuttle cab ride using Microcontroller 8051. We implemented the database using MySQL successfully and connected it to the Microcontroller 8051. RFID is successfully detecting the RFID tags with very low power consumption. It is a fully automated, reliable, transparent and convenient system. RFID cards being reusable are much more convenient compared to the paper-based ticketing system. These are used as universal travel pass cards that will allow any transportation on any route. GPS service along with the internet was used for distance measurement and fare calculation. GPS does not require the internet so is reliable even if there's no connectivity at all places of travel. Since fare calculation is done through the internet, the fare is crystal clear and provides no room for confusion. Database for travelers was created and accessed via the internet using a USB modem. Implementing this system can be seen as a step towards IoT. A powerful program algorithm can make the system publish real-time location data on the internet, ensuring the timekeeping of services. The system also holds bright promises towards different transportation fields, including transportation of school students ensuring a better level of safety

References

[1] P. A. Kamble and R. A. Vatti, "Bus tracking and monitoring using RFID," 2017 Fourth International Conference on Image Information Processing (ICIIP), Shimla, 2017
[2] K. G. Subadra, J. M. Begum and H. Dhivya, "Analysis of an automated bus tracking system for metropolitan using IoT," 2017 International Conference on Innovations in Information, Embedded and Communication Systems (ICIIECS), Coimbatore, 2017
[3] P. Y. Lau, Chu Qingxin and Wu Yueshan, "A new generation of smart UHF RFID reader," 2017 International Workshop on Electromagnetics: Applications and Student Innovation Competition, London, 2017
[4] International Journal of Advanced Research in Electrical, Electronics and Instrumentation Engineering (An ISO 3297: 2007 Certified Organization) Vol. 4, Issue 4, April 2015